

WorkflowGPT — Complete Project Documentation

Repository: <https://github.com/Ayush-2308/WorkflowGPT>

Application root: workflowgpt/

Local UI: <http://127.0.0.1:8000>

Document version: 2026-10-06

GitHub commit this doc describes: measured against local main (28 pytest passing)

This document is the full technical write-up of the project: purpose, architecture, how it was built, what is not included, how to run it with dummy request/response examples, and all project metrics that can be measured from the codebase.

1. One-line product definition

WorkflowGPT is a natural-language to n8n workflow agent: a FastAPI chat UI plus LangGraph pipeline that uses an LLM to plan (or ask for missing fields), generate n8n workflow JSON, deploy it to a self-hosted or cloud n8n instance, optionally test a webhook, and persist the run in Supabase.

It is not a WhatsApp/Instagram bot. Chat on port 8000 never sends a WhatsApp message. It only designs and deploys an n8n workflow; n8n executes automations after credentials exist there.

2. How it was built (engineering history)

Work proceeded in layers, matching git history:

Stage | What was added

Scaffold | Python package, Pydantic WorkflowSpec / PipelineState, .env

Parser | Anthropic (later OpenAI + Gemini) JSON extraction + one validation retry

Builder + deployer | Deterministic n8n node mapping + POST /api/v1/workflows + activate

Graph | LangGraph parse → build → deploy → test → persist

Completeness | HTTP/email/document/DB/IF mapping, failure persistence, list/get/delete APIs, pytest, CI

Agent | Conversational planner that can ask for tokens/IDs then emit wide n8n JSON

UI | Single-page chat at GET / calling POST /agent

Hardening | Gemini 503 retries, runpipeline import fix, .env never committed

External systems wired in: Gemini or Anthropic or OpenAI, n8n Public API, Supabase.

3. What exists vs what does not

Exists

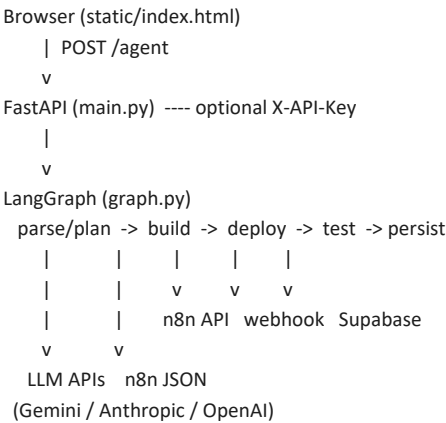
- Chat UI and REST API
- Multi-turn session (sessionid, answers, facts)
- Planner LLM → n8n JSON (catalog of common nodes + HTTP fallback)
- Fallback WorkflowSpec builder (webhook, schedule, form, HTTP, email, document, database, IF)
- n8n create / activate / delete / optional credential create
- Webhook or form test + execution poll
- Supabase workflows, executionlogs, agentsessions
- Health check of missing env names
- Unit tests (28) and GitHub Actions pytest

Does not exist

- Native “send me a WhatsApp right now” from the chat box
- Guaranteed support of every n8n community node
- OAuth login into Google/Slack/Facebook on behalf of the user
- Production Kubernetes / Terraform / full Docker app image (only optional n8n compose)
- Frontend framework (React/Next) — UI is one HTML file
- Multi-user auth / SSO (optional shared WORKFLOWGPTAPIKEY only)
- Horizontal scaling / Redis queue / worker pool

4. Architecture

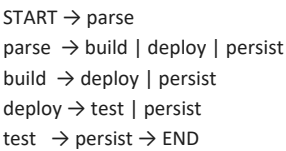
4.1 System context



4.2 Request path (chat)

1. Browser keeps workflowgpt-session in localStorage.
2. POST /agent with { message, sessionid, answers }.
3. db/sessions.py loads facts + last 40 messages (memory, then Supabase).
4. runpipeline runs the graph.
5. parsenode calls planturn (LLM, max 8192 tokens).
 - mode=ask → status needsinput, questions returned, no n8n deploy.
 - mode=build with n8nworkflow → sanitize, optional credentials, status built, skip deterministic builder, go to deploy.
 - workflowspec only → buildnode (builderagent).
 - LLM total failure → parser fallback; still failing → parsefailed.
6. deploynode POST /api/v1/workflows then activate. Missing n8n env → deployskipped.
7. testnode finds webhook (prefers Testwebhook) or form path; polls executions (20 × 0.75s).
8. persistnode upserts Supabase unless credentials missing. Successful test becomes completed.

4.3 LangGraph edges



Nodes: parse, build, deploy, test, persist (5).

5. Folder map

Path | Role

workflowgpt/main.py | FastAPI: UI, health, agent, CRUD
workflowgpt/graph.py | Pipeline orchestration
workflowgpt/config.py | Env load + missingsettings()
workflowgpt/schemas/models.py | Pydantic models
workflowgpt/agents/planneragent.py | Ask vs build JSON
workflowgpt/agents/parseragent.py | Spec parse + LLM HTTP + Gemini retries
workflowgpt/agents/builderagent.py | Spec → n8n nodes
workflowgpt/agents/n8nsanitize.py | LLM JSON cleanup
workflowgpt/agents/n8ncatalog.py | Node catalog text for the planner
workflowgpt/agents/deployeragent.py | n8n REST
workflowgpt/agents/executoragent.py | Test trigger + poll
workflowgpt/agents/expressions.py | Template → n8n expressions
workflowgpt/agents/credentialapply.py | Create/attach credentials
workflowgpt/db/supabaseclient.py | Run storage
workflowgpt/db/sessions.py | Chat sessions
workflowgpt/static/index.html | Chat UI
workflowgpt/migrations.sql | Schema
workflowgpt/tests/ | 9 files, 28 tests
.github/workflows/test.yml | CI pytest

6. Environment

Copy workflowgpt/.env.example to .env (gitignored).

Variable | Required to fully run | Notes
LLMAPIKEY | Yes | Provider secret
LLMPROVIDER | Default anthropic | gemini \ | anthropic \ | openai
LLMMODEL | Has defaults | e.g. gemini-3.8-flash
OPENAIBASEURL | OpenAI only | Default official API
GEMINIBASEURL | Gemini only | Default Google v1beta
SUPABASEURL | Persist | https://PROJECT.supabase.co
SUPABASEKEY | Persist | secret key, not publishable
N8NBASEURL | Deploy | No /api/v1 suffix
N8NAPIKEY | Deploy | Long JWT from n8n Settings → API
WORKFLOWGPTAPIKEY | Public hosts | Empty = open API

Gemini 429/503: up to 4 attempts with exponential backoff (2^attempt seconds).

7. How to start (local)

```
cd workflowgpt
python -m venv .venv
.\venv\Scripts\Activate.ps1
pip install -r requirements.txt
copy .env.example .env
# fill .env, run migrations.sql in Supabase
python -m uvicorn main:app --host 127.0.0.1 --port 8000
```

n8n must already listen on N8NBASEURL (often http://localhost:5678).

Browser: http://127.0.0.1:8000

OpenAPI: http://127.0.0.1:8000/docs

8. Dummy request / response (try these)

Replace 127.0.0.1:8000 if hosted elsewhere. Do not paste real secrets into GitHub issues.

8.1 Health

Request

```
GET /health HTTP/1.1
Host: 127.0.0.1:8000
```

Dummy response (configured)

```
{
  "status": "ok",
  "missing": [],
  "llmprovider": "gemini",
  "llmconfigured": true,
  "n8nconfigured": true,
  "supabaseconfigured": true,
  "authrequired": false
}
```

Dummy response (incomplete)

```
{
  "status": "incomplete",
  "missing": ["SUPABASEURL", "SUPABASEKEY", "N8NAPIKEY"],
  "llmprovider": "gemini",
  "llmconfigured": true,
  "n8nconfigured": false,
  "supabaseconfigured": false,
  "authrequired": false
}
```

8.2 Happy path — webhook to HTTP (use this first)

UI message (paste exactly)

When a webhook is received, POST the payload to <https://httpbin.org/post>.

Equivalent HTTP

```
POST /agent HTTP/1.1
Host: 127.0.0.1:8000
Content-Type: application/json
```

```
{
  "message": "When a webhook is received, POST the payload to https://httpbin.org/post."
}
```

Dummy success response (shape from a real local run)

```
{
  "requestid": "a1b7f0ed-c185-491c-889f-b1727cced659",
  "rawinstruction": "When a webhook is received, POST the payload to https://httpbin.org/post.",
  "deploymentstatus": "completed",
  "n8nworkflowid": "qDz8dpjzoMQ4PO5M",
  "n8nworkflowurl": "http://localhost:5678/workflow/qDz8dpjzoMQ4PO5M",
  "assistantmessage": "Deployed a webhook that forwards the body to httpbin.",
  "questions": [],
  "errors": [],
  "sessionid": "11111111-2222-3333-4444-555555555555",
  "testresult": {
    "status": "success",
    "webhook": {
      "url": "http://localhost:5678/webhook/workflowgpt-demo",
      "method": "POST"
    }
  }
}
```

```
}
```

What you should see in the UI: a green user bubble, then a bot message plus an n8n link. Opening the link shows Webhook → HTTP Request.

8.3 Agent asks for missing fields

Request

```
{
  "message": "Comment on new Facebook leads from Maharashtra."
}
```

Dummy needsinput response

```
{
  "deploymentstatus": "needsinput",
  "n8nworkflowid": null,
  "n8nworkflowurl": null,
  "assistantmessage": "I can build this in n8n, but I need access details.",
  "questions": [
    {
      "id": "pageid",
      "prompt": "What is the Facebook Page ID?",
      "secret": false,
      "placeholder": "123456789"
    },
    {
      "id": "pagetoken",
      "prompt": "Paste a Page access token.",
      "secret": true,
      "placeholder": ""
    }
  ],
  "errors": []
}
```

Follow-up (same session)

```
{
  "sessionid": "11111111-2222-3333-4444-555555555555",
  "message": "Here are the details you asked for.",
  "answers": {
    "pageid": "123456789",
    "pagetoken": "EAAB...redacted"
  }
}
```

Then the pipeline may build + deploy. Facebook Graph still needs a valid token; dummy values will deploy a workflow that fails at runtime.

8.4 Gemini busy (seen in production)

Same webhook prompt, LLM returns HTTP 503 twice (plan + retry).

Dummy error payload

```
{
  "deploymentstatus": "parsefailed",
  "assistantmessage": "I could not plan this automation yet.",
  "n8nworkflowurl": null,
  "errors": [
    "parse failed after validation retry: Gemini request failed with status 503: This model is currently experiencing high demand."
  ]
}
```

This is not a bad prompt. Wait and resend. Current code retries 503/429 up to four times per LLM call.

8.5 List runs

GET /workflows?limit=8 HTTP/1.1

```
{
  "workflows": [
    {
      "requestid": "a1b7f0ed-c185-491c-889f-b1727cced659",
      "rawinstruction": "When a webhook is received, POST the payload to https://httpbin.org/post.",
      "status": "completed",
      "n8nworkflowid": "qDz8dpjzoMQ4PO5M",
      "n8nworkflowurl": "http://localhost:5678/workflow/qDz8dpjzoMQ4PO5M"
    }
  ]
}
```

8.6 Delete

DELETE /workflows/a1b7f0ed-c185-491c-889f-b1727cced659 HTTP/1.1

```
{
  "requestid": "a1b7f0ed-c185-491c-889f-b1727cced659",
  "n8ndeleted": true
}
```

9. HTTP surface (complete)

Method	Path	Auth if key set	Purpose
--------	------	-----------------	---------

GET	/	No	Chat HTML
-----	---	----	-----------

GET	/health	No	Config flags
-----	---------	----	--------------

GET	/docs	No	FastAPI Swagger
-----	-------	----	-----------------

POST	/agent	Yes	Chat turn
------	--------	-----	-----------

POST	/generate-workflow	Yes	Same pipeline; instruction field
------	--------------------	-----	----------------------------------

GET	/workflows	Yes	List
-----	------------	-----	------

GET	/workflows/{requestid}	Yes	Detail
-----	------------------------	-----	--------

DELETE	/workflows/{requestid}	Yes	n8n + DB delete
--------	------------------------	-----	-----------------

Count: 7 application routes (+ /docs, /openapi.json from FastAPI).

Auth header: X-API-Key or Authorization: Bearer.

10. Data models and database

PipelineState (API + graph)

requestid, rawinstruction, workflowspec, n8nworkflowjson, n8nworkflowid, deploymentstatus, testresult, errors, assistantmessage, questions, facts (secrets redacted in HTTP as •••• when the key name looks secret), conversation, sessionid.

WorkflowSpec (builder fallback)

name, description, trigger{type,config}, actions[{type,config,dependson}], rawinstruction.

Tables (migrations.sql)

Table	Keys / notes
-------	--------------

workflows	PK requestid; JSON spec + n8n JSON; status; errors
-----------	--

executionlogs	FK requestid ON DELETE CASCADE
---------------	--------------------------------

agentsessions	PK sessionid; facts (may hold tokens); messages
---------------	---

RLS enabled with permissive service-role policies.

11. Status dictionary

Status | Meaning

pending | Graph start

needsinput | Waiting on user questions

parsed | Spec ready, builder next

built | n8n JSON ready

deployed | n8n id assigned

tested | Test finished (then stored as completed)

completed | Successful test persisted

parsefailed | Plan/parse failed

buildfailed | JSON/spec mapping failed

deploymentfailed | n8n API error

deployskipped | n8n env empty

testfailed | Trigger/poll failed

testskipped | No webhook/form to hit

storefailed | Supabase write failed

error | logerror crash path

12. Technical metrics (measured 2026-10-06)

Figures exclude nodemodules, .venv, .tools, pycache, .git.

Size and structure

Metric | Value

Python files | 29 (20 application, 9 test)

Python raw lines | 3773 (src 3250, tests 523)

Python LOC (non-blank, non-#) | 3292 (src 2850, tests 442)

Chat UI HTML lines | 228

Markdown lines (repo docs before this refresh) | ~521

Tracked text-ish source files counted | 40

Agent modules under agents/ | 10 Python files

Graph nodes | 5

Conditional edge groups | 3

HTTP app routes | 7

LLM providers implemented | 3

Env keys in .env.example | 10

Supabase tables | 3

pytest files | 9

pytest tests | 28 passed (runtime ~11.4 s on this machine)

GitHub Actions jobs | 1 (pytest on Python 3.12)

Documented git commits on main (at measure) | 6

Runtime / limits (from code)

Metric | Value

FastAPI bind (docs) | 127.0.0.1:8000 local; 0.0.0.0:8000 for hosts

Default n8n | http://localhost:5678

LLM HTTP timeout | 60 s

n8n HTTP timeout | 30 s

Gemini max output (chat) | 4096 default; planner 8192
Gemini transient retries | 4 (status 429, 503)
Parser validation retries | 1 extra LLM call
Execution poll | 20 attempts $\times$ 0.75 s $\approx$ 15 s
Session messages kept | last 40
Workflow list cap | 1–200 (default 50)
Health missing | names only, never secret values

Test breakdown (28)

File | Tests
testapi.py | 4
testbuilder.py | 4
testdeployer.py | 2
testexecutor.py | 4
testexpressions.py | 2
testgraph.py | 3
testparser.py | 5
testplanner.py | 2
testsanitize.py | 2

Coverage style: unit tests with mocks; no live Gemini/n8n in CI.

Builder catalog (deterministic path)

Triggers: webhook, schedule, formsubmission.

Actions: httprequest, sendemail, generatedocument, databaseinsert (+ IF on condition).

Planner catalog additionally steers toward Slack, Gmail, Sheets, Telegram, Postgres, HTTP, Code, Switch, etc.

Dependencies (requirements.txt)

fastapi, uvicorn, langgraph, pydantic, python-dotenv, supabase, httpx, anthropic, pytest (minimum versions pinned with $\geq$).

No React, no Redis, no Celery.

13. Security metrics / practices

Item | Behavior
.env in git | Ignored
Health endpoint | Does not echo secrets
API facts | Keys matching token/secret/password/key/authorization/cookie $\rightarrow$ ••••
Public deploy | Set WORKFLOWGPTAPIKEY
Supabase key | Server sbsecret, never sbpublishable
n8n key | JWT; short UI ids are rejected (401)
Session facts | Stored in DB — treat Supabase as sensitive

14. Deploy notes

Hosting the GitHub repo is not a running product. You must run WorkflowGPT and an n8n the API can reach.

- Same machine: uvicorn + local n8n.
- Internet: Render/Railway for the API and n8n Cloud/VPS. N8NBASEURL must be the public n8n URL. Cloud API + laptop localhost:5678 will fail deploy.

Start command from workflowgpt/:

uvicorn main:app --host 0.0.0.0 --port 8000

15. Cross-check (documentation vs code)

Verified against the tree on 2026-10-06:

Claim | Evidence

Chat UI at / | main.py home() reads static/index.html

Agent POST /agent | main.py agentturn

Graph 5 nodes | buildgraph() in graph.py

28 tests pass | pytest -q → 28 passed in 11.41s

7 app routes | @app.get/post/delete in main.py

3 LLM providers | callllm branches in parseragent.py

3 SQL tables | migrations.sql

Gemini retry | callgemini loop range(4)

Secrets not in git | .gitignore contains .env

Planner ask/build | agents/planneragent.py AgentPlan.mode

Dummy happy path used in real demo | Local deploy ids such as qDz8dpjzoMQ4PO5M were created on this machine; ids on your n8n will differ

If a later commit adds routes or tests, re-run pytest and update section 12.

16. Troubleshooting

Symptom | Action

parsefailed + Gemini 503 | Wait, retry same webhook prompt

runpipeline is not defined | Restart uvicorn after pull

n8n 401 | Use full API JWT, not list id

Empty history | Run migrations.sql; check Supabase keys

WhatsApp not arriving | Expected — app does not send WhatsApp

17. License / data ownership

Source: GitHub Ayush-2308/WorkflowGPT.

Runtime secrets and workflow data live in the operator's .env, n8n, and Supabase — not in the public repo.